MIL-C-1997C
11 April 1983
SUPERSEDING
MIL-C-1997B
11 October 1967

MILITARY SPECIFICATION

COVER, WATERPROOF, MULTIPURPOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of waterproof cover.

1.2 Classification. The cover shall be of the following sizes as specified (see 6.2):

Size 1 - 8 inches wide, 18 inches long.
Size 2 - 10 inches wide, 56 inches long.
Size 3 - 15 inches wide, 45 inches long.
Size 4 - 20 inches wide, 84 inches long.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
T-C-571	-	Cords, Cotton; General and Special Purpose, Sash and Venetian Blind
UU-P-268	-	Paper, Kraft, Wrapping
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8465

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STANDARDS

FEDERAL

FED-STD-406 - Plastics: Methods of Testing
FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

* 3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2 and 6.3).

3.2 Materials.

* 3.2.1 Film. The film used in fabricating all covers shall be polyethylene lay flat tubing conforming to type I, grade B, finish 2 of L-P-378, except that the workmanship requirements are not applicable. The density of the film shall be 0.914 to 0.950 grams per cubic centimeter, and the film shall have a minimum impact resistance of 125 grams. The film shall be 0.004 inch thick with a tolerance of ± 25 percent and the thickness shall apply to the individual determination. The color of the film shall approximate color No. 24087 of FED-STD-595.

3.2.2 Cord. The cotton cord shall conform to type I, class 3, size No. 4 of T-C-571. The color of the cord shall be Olive Green 107.

* 3.3 Construction. All covers shall be constructed from extruded lay flat tubing of the polyethylene specified in 3.2.1. One end of the tube shall be closed with a heat seal not less than 1/16 inch wide nor more than 3/4 inch wide. Excess material beyond the seal shall be trimmed evenly and shall not exceed 1/2 inch. Each size 2 and size 3 cover shall be furnished with a 72 ± 2 inch piece or the cotton cord specified in 3.2.2. The ends of the cord shall be either tipped or resin coated. The cord shall not be attached to the cover (see 5.1.1.2).

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3.3.1 Label. Each cover shall be labeled in accordance with type IV, class 5 of DDD-L-20 except that the fastness of printing test shall not apply. The label shall be clearly legible after being subjected to the ink adhesion test specified in 4.4.4. The label shall be located midway (± 2 inches) between the sealed seam and the opening.

3.4 Dimensions. The inside finished measurements and tolerances for each size cover shall be as specified below:

	Width (inches)	Length (inches)
Size 1	8 ± 1/4	18 ± 1/2
Size 2	10 ± 1/4	56 ± 1
Size 3	15 ± 1/2	45 ± 1
Size 4	20 ± 5/8	84 ± 1-1/2

3.5 Performance.

3.5.1 Seam length. The breaking strength of the seam shall be not less than 60 percent of the breaking strength of the film when tested as specified in 4.4.4.

3.5.2 Crease strength. The breaking strength at the side creases of the cover shall be not less than 50 percent of the breaking strength of the film when tested as specified in 4.4.4.

3.5.3 Leakage. The cover shall not leak or separate at the seam when tested as specified in 4.4.4.

* 3.6 Workmanship. The finished covers shall be clean, well finished, free from gels, pinholes, holes, cracks, blisters, fish eyes, mottling, particles of foreign matter and undispersed raw material. Raw edges shall be free from nicks and cuts. The quality of the covers shall conform to the levels established by this document.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3 and tested for the characteristics specified in table II. The presence of any defect failure to pass any test shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document.

* 4.4.2 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of one cover. The sample unit shall be one cover. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

TABLE I. Visual defects

Examine	Defect
Polyethylene film	Not approximately color No. 24087. Not uniform in color, texture and finish. (Color difference due to variation in film thickness shall be excluded.) Any gel, pinhole, hole, crack, blister, fish eye, undispersed raw material, mottling. Any foreign matter. Dirty.
Cotton cord (when applicable)	Not color OG 107. Not one piece. Any cut. Chafed or damaged. Core missing. Kinks, broken or loose ends. Unevenly braided resulting in open places, breaks in continuity of braid or soft spots. Finish gummy or tacky. Ends not tipped or resin coated.

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TABLE I. Visual defects (cont'd)

Examine	Defect
Label	Label missing. Impression of label too heavy, objectionable offset or strike through. Label not as specified. Label improperly located.
Construction and workmanship	Not constructed from lay flat tubing. Varies from specified design and construction. Not heat sealed across one end. Any area of seal thin or not continuous. Poor heat seal resulting in inadequate fusion. Excess material not trimmed evenly. Any nick or cut in raw edge. Cord missing (sizes 2 and 3). Cord attached to cover (sizes 2 and 3).

* 4.4.3 End item dimensional examination. The end item shall be examined for the defects listed below. The lot size shall be expressed in units of one cover. The sample unit shall be one cover. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

Examine	Defect	
	More than	Less than
	Inches	Inches
Size 1 - Width (inside)	8-1/4	7-3/4
- Length (inside)	18-1/2	17-1/2
Size 2 - Width (inside)	10-1/4	9-3/4
- Length (inside)	57	55
Size 3 - Width (inside)	15-1/2	14-1/2
- Length (inside)	46	44
Size 4 - Width (inside)	20-5/8	19-3/8
- Length (inside)	85-1/2	82-1/2
Heat seal - Width	3/4	1/16
- Excess material beyond seal	1/2	---
Cord length (sizes 2 and 3)	74	70

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* 4.4.4 End item testing. Each end item lot shall be tested for the characteristics listed in table II. The sample unit shall be three covers (one for the label legibility test, one for the seam and crease strength tests, and one for the leakage test). The sample shall consist of three sample units drawn at random from the lot. Any test result failing to conform to the specified requirement shall be cause for rejection of the lot.

TABLE II. End item tests

Characteristic	Requirement paragraph	Test method paragraph
Label adhesion	3.3.1	4.5.1
Seam strength	3.5.1	4.5.2
Crease strength	3.5.2	4.5.2
Leakage	3.5.3	4.5.3

* 4.4.5 Packaging inspection. An examination shall be made to determine that preservation, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged. Examination of closure defects listed below shall be made on shipping containers fully packaged. The lot shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as inadequate tying of cord bundles. Bulged or distorted container.
Content required.	Number of stacks per container is more or less than Average number of covers per stack is more or less than required. 1/

1/ For this defect, one stack from each ten stacks (or fraction thereof) within the shipping container shall be examined.

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* 4.5 Methods of inspection.

* 4.5.1 Ink adhesion (label) test. The ink adhesion shall be determined according to the procedure in the ink adhesion test of L-P-378 except that the test shall be performed on the label of a finished cover. The test area of the label shall be visually examined for legibility.

* 4.5.2 Seam and crease strength tests. Strength of the heat sealed seam area and the creased area shall be determined in accordance with method 1013 of FED-STD-406. The specimens shall be 1 inch in width and shall be cut from a finished cover with the seam or crease across the short dimension and equidistant from the ends. The gage marks need not be applied and the elongation need not be measured. The initial separation between the grips of the apparatus shall be 2 inches and the specimen shall be placed in grips with the seam or crease equidistant between the grips. The strength of the seam or crease reported in pounds to the nearest 0.1 pound, shall be the average of 5 determinations per sample unit. The film strength shall be determined by the same method used to determine the seam strength except that the specimens shall be cut from the unseamed and uncreased section of the cover with the long dimension parallel to the long dimension of the cover and all reference to the seam or crease shall not apply. The strength of the film, reported in pounds to the nearest 0.1 pound, shall be the average of 5 determinations per sample unit.

* 4.5.3 Leakage test. The cover shall be marked 5 inches from the inside of the seam and filled with water to the mark. The temperature of the water shall be 73.5 + 2°F. The cover shall then be suspended over a dry receptacle for a period of 3 hours. At the end of the 3-hour period, the cover shall be examined for seam separation and leakage of water into the receptacle.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Covers. Covers of one size only, shall be neatly stacked in the quantities and folded to the dimensions specified in table I.

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TABLE III. Covers

Size of cover (inches)	Quantity per stack	Folding instructions	Approximate dimensions of folded stacks (inches)	Quantity per shipping container
Size 1 (8 x 18)	100	No folding required	18 x 8	1000
Size 2 (10 x 56)	25	Stacks shall be folded in thirds the length	19 x 10	350
Size 3 (15 x 45)	50	Stacks shall be folded in half the length	24 x 15	250
Size 4 (20 x 84)	25	Stacks shall be folded twice in half the length	21-1/2 x 20	100

5.1.1.2 Cords, cotton. Cords shall be bundled in the quantities specified in table II and securely tied with cotton cord. The bundles shall be of proper size to fit the shipping container.

TABLE IV. Cords

Size of covers (inches)	Quantity per bundle	Quantity per shipping container
Size 1 (8 x 18)	0	0
Size 2 (10 x 56)	25	350
Size 3 (15 x 45)	50	250
Size 4 (20 x 84)	0	0

* 5.1.2 Commercial. Covers and cords shall be preserved in accordance with MIL-STD-1188.

* 5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

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* 5.2.1 Level A packing. Covers of one size only, preserved as specified in 5.1, shall be packed in quantities specified in table I within a snug-fitting fiberboard shipping container conforming to style RSC-L grade V2s of PPP-B-636 except that the top and bottom flaps shall be glued. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Cords, perserved as specified in 5.1, shall be packed flat on top of size 2 and size 3 covers in the quantities specified in table II. Each fiberboard container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. In addition, size 1 covers shall have a full size slip sheet of the above specified kraft paper placed between each stack. Each container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636.

* 5.2.2 Level B packing. Covers of one size only, preserved as specified in 5.1, shall be packed in the quantities specified in table I within a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF, (variety SW) or SF, class domestic, grade 275, of PPP-B-636, except that top and bottom flaps shall be glued. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Cords preserved as specified in 5.1, shall be packed flat on top of size 2 and size 3 covers in the quantities specified in table II. Each fiberboard container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. In addition, size 1 covers shall have a full size slip sheet of the above specified kraft paper placed between each stack. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

* 5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

* 5.2.3 Commercial packing. Covers and cords, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

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6. NOTES

6.1 Intended use. The items covered by this document are intended to provide waterproof, lightweight protection for small arms against rain and moisture. The covers are used for the following purposes:

Size 1 - For pistol or personal effects.
Size 2 - For rifle or carbine.
Size 3 - For sub-machine gun.
Size 4 - For machine gun.

The covers may also be used as water carrying bags, or protectors for miscellaneous small items of clothing and equipment, or they may be inflated to improvise lightweight water crossing rafts.

* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this document.
(b) Size required (see 1.2).
(c) When a first article is required (see 3.1, 4.3 and 6.3).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

* 6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of one cover. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

* 6.4 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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